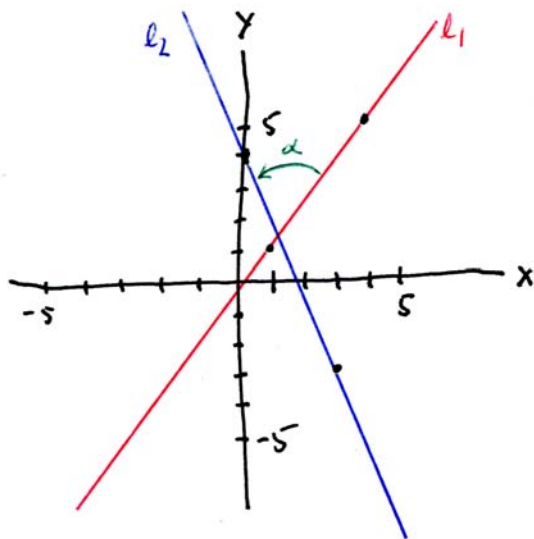


Name \_\_\_\_\_

Math 1350, Quiz #3

1. (Section 1.6, Problem 11) Find the tangent of the angle from line  $l_1$  to  $l_2$ , where  $l_1$  contains the points (4,5) & (1,1) and  $l_2$  contains the points (3,-3) & (0,4).



$$m_1 = \frac{1-5}{1-4} = \frac{4}{3} \quad m_2 = \frac{4-3}{0-3} = -\frac{7}{3}$$

$$\tan \alpha = \frac{-\frac{7}{3} - \frac{4}{3}}{1 + \frac{7}{3} \cdot \frac{4}{3}} = \frac{-\frac{11}{3}}{1 - \frac{28}{9}}$$

$$= \frac{-\frac{11}{3}}{-\frac{19}{9}} = \frac{33}{19}$$

2. (Section 1.7, Problem 15) Find the points of intersection and sketch the graphs of the equations

$$\begin{cases} 3x + 2y = -1 \\ x - 4y = -12 \end{cases}$$

$$\begin{array}{r} 6x + 4y = -2 \\ x - 4y = -12 \\ \hline 7x = -14 \Rightarrow x = -2 \end{array}$$

$$\begin{array}{r} -2 - 4y = -12 \\ -4y = -10 \Rightarrow y = \frac{5}{2} \end{array}$$

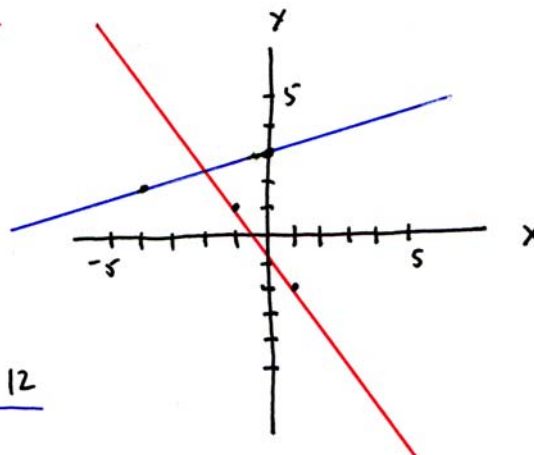
$$\left(-2, \frac{5}{2}\right)$$

$$3x + 2y = -1$$

| x  | y  |
|----|----|
| 1  | -2 |
| -1 | 1  |

$$x - 4y = -12$$

| x  | y |
|----|---|
| -4 | 2 |
| 0  | 3 |



3. (Section 2.1, Problem 13) Find a unit vector in the direction of  $\mathbf{v} = \mathbf{i} + 2\mathbf{j}$ .

$$|\vec{v}| = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$\vec{u} = \frac{1}{\sqrt{5}} (\vec{i} + 2\vec{j}) = \frac{1}{\sqrt{5}} \vec{i} + \frac{2}{\sqrt{5}} \vec{j}$$

4. (Section 2.1, Problem 25) Find  $\mathbf{u} + \mathbf{v}$ ,  $\mathbf{u} - \mathbf{v}$  and draw a diagram showing  $\mathbf{u}$ ,  $\mathbf{v}$ ,  $\mathbf{u} + \mathbf{v}$ ,  $\mathbf{u} - \mathbf{v}$  where  $\mathbf{u} = 3\mathbf{i} - \mathbf{j}$ ,  $\mathbf{v} = \mathbf{i} + 2\mathbf{j}$ .

$$\vec{u} + \vec{v} = 4\vec{i} + \vec{j}$$

$$\vec{u} - \vec{v} = 2\vec{i} - 3\vec{j}$$

